MIL-C-24643/19B
3 October 1986
SUPERSEDING
MIL-C-0024643/19A(SH)
8 November 1985
MIL-C-24643/19
28 September 1984

MILITARY SPECIFICATION SHEET

# CABLE, ELECTRICAL, 1000 VOLTS, TYPE LS6SGU (INCLUDING VARIATION LS6SGA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Copper conductor, uncoated, class D stranding (see table I for size).
Second - Silicone rubber - glass tape insulation (see table I for diameter). Letter identification code applied by method 5 (two conductors each of A, B and C).
Third - The six conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used to form a firm, well-rounded assembly. Cabling sequence shall be A, B and C.
Fourth - Binder or combination binder and barrier.

Construction variant for type LS6SGU

Fifth - Cross-linked polyolefin jacket. (See table I for thickness.) Cable surface marking required.

Construction variant for type LS6SGA

(B) Fifth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking not required.
Sixth - Braided metal armor and paint.

(B) denotes changes.

AMSC N/A FSC 6145
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited



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TABLE I. Details.

| Military 1/ part no. M24643/19 | Size 2/ | Conductor size AWG | Type's overall diameter (inches) | | | | Diameter over insulation nominal (inch) | Jacket thickness (nominal) (inch) | Conductor resistance per 1000 feet at 25°C maximum (ohms) | Insulation resistance per 1000 feet minimum (megohms) | Cold bending mandrel diameter maximum (inches) | Accel-erated service loading amperes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | LS6SGU | | LS6SGA | | | | | | | |
| | | | min | max | min | max | | | | | | |
| -01UN | 100 | 0(class D) | --- | 1.600 | --- | 1.650 | 0.453 | 0.060 | 0.106 | 160 | 20 | 475 |
| -02UN | 125 | 00(class D) | --- | 1.790 | --- | 1.840 | .507 | .060 | .0850 | 150 | 22 | 520 |
| -03UN | 150 | 000(class D) | --- | 1.960 | --- | 2.010 | .557 | .060 | .0674 | 135 | 24 | 570 |
| -04UN | 200 | 0000(class D) | --- | 2.200 | --- | 2.250 | .634 | .075 | .053 | 125 | 27 | 650 |

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS6SGU and LS6SGA.



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EXAMINATION AND TESTS:

| | Requirements |
|---|---|
| **Basic electrical:** | |
| Conductor resistance - ohms/1000 feet, at 25°C, maximum. | (see table I) |
| Voltage withstand - volts, root mean square, minimum.... | 5000 |
| Insulation resistance - megohms/1000 feet, minimum..... | (see table I) |
| Conductor continuity..... | No failure |
| **Group A:** | |
| Visual and dimensional..... | No failure |
| Watertightness - see MIL-C-24643 for limits of water leakage..... | No failure |
| **Group B:** | |
| Gas flame - 1 hour..... | No failure |
| Drip - 95 ± 1°C..... | Zero |
| Tear - pounds per inch thickness, minimum (ASTM D 470).. | 35 |
| Cold bending, cable - (see table I for mandrel diameter) | No damage |
| Physicals (unaged) | |
| Jacket (cable) | |
| Tensile strength - $lbf/in^2$, minimum..... | 1300 |
| Elongation - percent, minimum..... | 160 |
| **Group C:** | |
| Physicals (aged) | |
| Jacket (cable) | |
| Air oven | |
| Tensile strength - percent of unaged, minimum..... | 60 |
| Elongation - percent of unaged, minimum.. | 60 |
| Hot oil immersion | |
| Tensile strength - $lb/in^2$, minimum..... | 50 |
| Elongation - percent, minimum..... | 50 |
| Shrinkage..... | No failure |
| Heat distortion - percent of unaged, maximum..... | 30 |
| Permanence of printing (jacket) - cycles, minimum..... | 250 |
| Cable filler removability..... | No failure |
| Armor (type LS6SGA only) - conformance to material, construction and coverage..... | No failure |

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

| | |
|---|---|
| Cold working (minus 20 ± 2°C)..... | No damage |
| Gas flame (3 hours)..... | No failure |
| Accelerated service - all sizes (see table I for load current)..... | No failure |
| Aging and compatibility (cable) (125 ± 5°C)..... | No failure |
| (B) Abrasion resistance (jacket) - scrapes, minimum..... | 75 |



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QUALIFICATION INSPECTION: (Continued)

| | Requirements |
|---|---|
| Acid gas equivalent - percent, maximum | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Halogen content - percent, maximum | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Flame propagation (cable) | No failure |
| Immersion (jacket) | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Smoke index, maximum | |
| Jacket | 25 |
| Fillers | -- |
| (B) Insulation | 35 |
| Toxicity index, maximum | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) - (type A) hardness, minimum | 80 |
| Weathering (jacket) | No failure |

UNIT ORDERING LENGTHS:
All sizes 800 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-19)

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